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Low-Density Steels: Complex Metallurgy for Automotive Applications

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The current aim in the development of third-generation steels for lightweighting automotive applications is to increase strength keeping at least the same formability as current steel concepts. In this philosophy, an optimal concept would be one that brings, in addition, a lower density. For this purpose, low-density steels have been designed with important aluminum additions obtaining density reductions of 8–10% or higher in comparison with low-carbon steels. At the levels required for lightweighting, aluminum introduces complex phenomena in steels. Here, some of the effects of aluminum in phase stability, CALPHAD-type modeling, and microstructure development are described, the latter in relation with mechanical properties. Finally, the potential of two families of lightweight steels for automotive applications is assessed by comparison with a steel currently present in automotive structures.

ALUMINUM: AN "OLD" ALLOYING ELEMENT

Aluminum is well known in steel production as a deoxidizer. It was used originally to "quiet" over-oxidized heats, but its use as a deoxidizer became widespread before World War II as considerable data showed its effectiveness in, for instance, controlling grain size. A thorough description of this early use can be found in Ref. 1 Before that, in steelmaking practice, aluminum was a sign of low-quality steel. The following statement shows the reputation of aluminum in 1936:

"In Sheffield, they had always been led to regard the doping of steel with aluminum as a subterfuge of those who were not experts in the science of making steel. In other words, if a steel melter required to use a lot of aluminium to make the steel sound, he hardly merited the outstanding position which he occupied."[1]

It Ref. 1 where there is also a description of the different types of inclusions present in steel when it is deoxidized with aluminum, e.g., oxide, sulfide, and nitride inclusions. Furthermore, the effects on grain size, notch sensitivity, and aging are also reviewed.

In spite of its bad reputation in steelmaking, other sources of information showed earlier evidence of its use as alloying element where the distinguished Robert A. Hadfield[2] patented a group of steels in 1887 containing from 0.1 to 20 Al wt.%. A few years later, he described experiments on the effect of aluminum in iron-carbon alloys with contents up to 5.6 wt.%.[3] For his research, a material with an even higher aluminum content (9.14 wt.%) was produced but not studied. That was the time when the concept of the allotropic forms of iron had been recently proposed by the great French metallurgist Floris Osmond, William C. Roberts-Austen had not yet published his first Fe-C phase diagram and the discussion and establishment of the now well-known terms ferrite, pearlite, etc. was just starting. The latter terms were from the recent

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proposal (at the time) of Henry M. Howe in his celebrated *The Metallurgy of Steel* published in 1890.

In this context, the conclusions of Hadfield[3] were focused not on the effect on structure but on the effect on fluidity, forging, fracture tests, etc. Among those results, Hadfield proved with the aid of Osmond that aluminum does not affect the melting point of iron as much as it was discussed at the time. A reduction of 25°C for an alloy including 5.6 wt.% aluminum was mentioned.

In summary, Hadfield showed that the effect of aluminum was favorable but slight in tensile testing and none on hardness testing. These statements, as previously mentioned, were done without any indication of the structure of the samples. Hadfield recognized the similarity with silicon concerning its effects but stated that economic factors will hinder the application of aluminum even at the small quantities he recommended (0.1–0.15%). Hadfield's paper was well received and generated an extensive discussion from many, including Osmond and Le Châtelier. The discussion showed that aluminum was being seriously studied as an alloying element for steels. Osmond discussed, probably for the first time, the important influence of aluminum as a ferrite stabilizer by saying, "If the curve of cooling of the alloy containing 5% aluminum is examined, and compared with that of an extra soft steel containing nearly the same percentage of carbon, it will be seen that the point ar3 is wanting. The aluminum thus maintains the iron, for the most part, in the α-state from the ordinary temperature to the melting-point."

After the establishment of the importance of phase diagrams and a more unanimous steel nomenclature, later investigations in high aluminum-added steels dealt with the identification of the different microstructural components,[4–7] the determination of phase stability,[4,8–11] and mechanical properties mainly in high-manganese alloys.[12–14] Literature on this subject significantly increased after 1980 when replacement of Cr-Ni stainless steel by using austenitic alloys of the Fe-Mn-Al-C system was forecasted.[15] In this period, research was also conducted first on microstructure evolution in rapidly solidified alloys[16] and later on two-phase austenite-ferrite microstructures.[17] Interest in aluminum for the development of steels with intermetallics was also envisaged.

Nevertheless, the effect of aluminum additions on density reduction of iron alloys was not mentioned, to the author's knowledge, until Frommeyer et al.[18] from Max-Planck-Institut für Eisenforschung GmbH (MPIE) discussed in the late 1990s the reduction in density as a function of aluminum content for binary Fe-Al alloys (Fig. 1). This effect, discussed in the next section, has become, since then, an important design parameter in the development of steels for lightweighting automotive applications due to strict European Union regulations concerning CO_2 emissions for the coming years

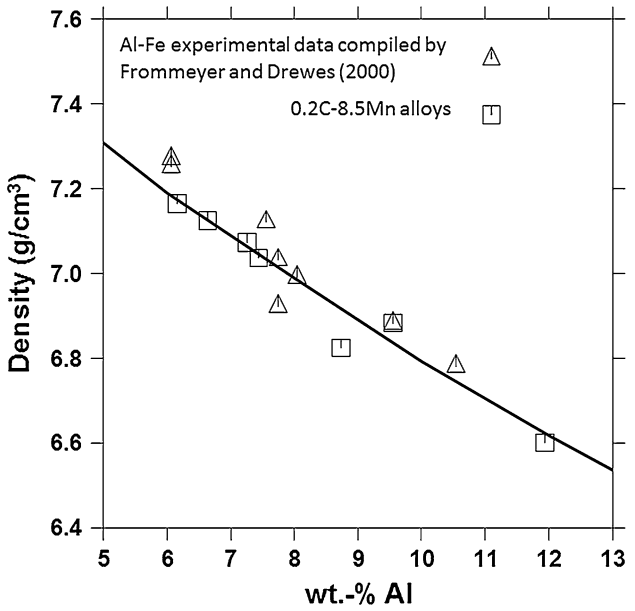


Fig. 1. Density as a function of aluminum content for binary Fe-Al[18] and for quaternary 0.2C-8.5Mn (wt.%) alloys studied in this work. The latter were evaluated by picnometry. The continuous line corresponds to the density calculation for the quaternary system using Thermo-Calc (see text for explanation).

and alternative and competitive materials. Several new alternative concepts like carbide-free bainite, medium manganese steels, etc. could also bring an important reduction in the weight of vehicles because of the important levels of strength and elongation obtained. However, there is a hidden limit dictated by structural rigidity concepts. Therefore, an optimal concept would be one that brings in addition to an increased strength at iso-formability a lower density, allowing car makers to use higher thicknesses.

EFFECT OF AL ON STEEL DENSITY

This effect results from the combination of the lower atomic mass of aluminum and the effect aluminum has in expanding the lattice parameter of iron alloys. Even though higher additions of aluminum could be desirable for weight reduction, ordered phases will appear at a certain level of aluminum (B2 and/or DO_3). This phenomenon will limit the amount of aluminum because the presence of these ordered phases will render the material brittle.

The experimental data (Fig. 1) seems to indicate a slightly lower density of quaternary alloys in comparison to the binary Fe-Al. There is probably an effect of Mn, but in this case the two data come from two different sources, which might also explain the difference. The Thermo-Calc calculation (Thermo-Calc Software, Inc., McMurray, PA, USA) was performed using the database with the latest assessment by Lindahl and Selleby[19] with volume data from the commercial database Thermo-Calc

Software TCFE7 Steels/Fe-alloys database version 7.[20] That assessment will be further described in the next section. In this calculation, the phase fractions of austenite and ferrite were calculated in the same range of temperatures used for preparing the experimental alloys. The density was then evaluated with these phase fractions at room temperature. Agreement is excellent at the lower aluminum contents, whereas at higher aluminium contents the κ-phase, which is not considered in this calculation, might be present at these temperatures and could explain the difference with the experimental data.

INCORPORATING ALUMINUM IN FE-C THERMODYNAMIC CALCULATIONS

Aluminum as a strong ferrite stabilizer makes binary Fe-Al alloys basically ferritic, and when added in sufficient quantities (Al > 6 wt.%) to low carbon steels, the matrix will be ferritic. The simultaneous addition of C and Mn leads to the presence and stabilization of austenite down to low temperatures. Aluminum additions bring, besides order–disorder transformations, another complex metallurgical phenomenon to steels: spinodal decomposition. This has been observed in the quarternary FeMnAlC system,[21,22] but it should also occur in the ternary FeAlC system.[7]

Another particularity of the addition of high levels of Al is the replacement of cementite by an ordered carbide, the κ-phase. This phase has a perovskite structure, designated as $E2_1$ in the Strukturbericht classification, but with an uncompleted occupancy of the C atom. The formula commonly used is the $(Fe,Mn)_3AlC_x$ even though results from the literature differ from this formula, mainly in the case of the ternary system Fe-Al-C and the relation between Fe and Al. To tailor the microstructures and because of the multiple combinations possible, thermodynamic calculations are required. This has not been possible until recently because of the lack of a proper thermodynamic database that takes into consideration the effect of large additions of Al.

The incorporation of CALPHAD-type modeling into the study of phase equilibria in the ternary Fe-Al-C started with the work of Hari Kumar and Raghavan.[23] They applied the compound energy formalism[24] for the solution phases. Even though they recognized that the κ-phase presents a range of compositions they used an average fixed composition of $Fe_3AlC_{0.565}$. They produced a liquidus projection and eight isothermal sections together with a complete reaction scheme.

Ohtani et al.[25] introduced first-principles calculations into CALPHAD modeling to calculate the formation energy of the κ-phase (Fe_3AlC) and proceeded with the CALPHAD approach taking as experimental model the work of Palm and Inden.[11] A model with two substitutional sublattices and one interstitial sublattice was used for the description of the ordered phases for the body-centered cubic (bcc) structures and the κ-phase was modeled with another model also with two substitutional sublattices. In this case, the κ-phase was treated as a solid-solution phase. Comparing the calculated energies of formation from the κ-phase and the related $L1_2$, they determined that interstitial carbon enhances the stability of the $L1_2$ structure. Their thermodynamic modeling of the Fe-Al-C phase diagram is in accordance with the experimental work by Palm and Inden.[11]

Connétable et al.[26] did not find the thermodynamic model in Ref. 25 satisfactory and reassessed the Fe-Al-C system. They proposed a new CALPHAD description of the κ-phase based on their order–disorder model for the face-centered cubic (fcc) phase with four substitutional sublattices but restricted the carbon contents to 20 mol.% C. together with new ab initio calculations, the latter described elsewhere.[27] The fcc and bcc interactions in the Al-C system were revised using experimental results from the ternary system,[11] something that was not done before. In the Al-Fe system, the fcc description (γ-loop) was improved with the help of an ordering contribution. These revisions improved the ternary description of the fcc and bcc phases where they used the experimental tie lines given in Ref. 11 at 1200°C as a guide. The comparison of the modeling with the experimental isothermal sections was satisfactory and in the case of the isopleths was very good. The rather large solubility of carbon in ferrite shown at 800°C and 1000°C in Ref. 11 was assumed to have been caused by inaccuracies in the electron probe microanalysis measurements and is not reproduced by the model. The Al-Fe system was further revised by Sundman et al.[28] using a four-sublattice model for the bcc phase (in addition to the already modeled fcc phase) and using the most recent experimental data.

A more complete database has been recently built by Hallstedt in the frame of a Research Fund for Coal and Steel (RFCS) project.[29] In a first version, it incorporated the assessment of the FeMnAlC system by Chin et al.[30] and several sources from the literature. This assessment[30] includes the Fe-Al-C description from Ref. 26 but with a simplified model for the κ-phase. It is no longer described as an ordered branch of the fcc phase but as a separate phase with a fixed (Fe + Mn)/Al ratio. They also modeled the Mn-Al-C system, which also contains the κ-phase. A second version of this database included the work of Hallstedt and coworkers on the Mn-C[31] and the Fe-Mn-C[32] systems. In the Fe-Mn-C system, fcc-carbide equilibria at a low temperature and melting equilibria were improved. In this case, the Al-Fe-Mn description was that of Umino et al.[33] In the same RFCS project, Lindahl and Selleby[19] have very recently improved the Al-Fe-Mn system description by using the latest assessments of the Al-Fe and Al-Mn binaries. The complete system, including extensive data on fcc-bcc equilibria in the

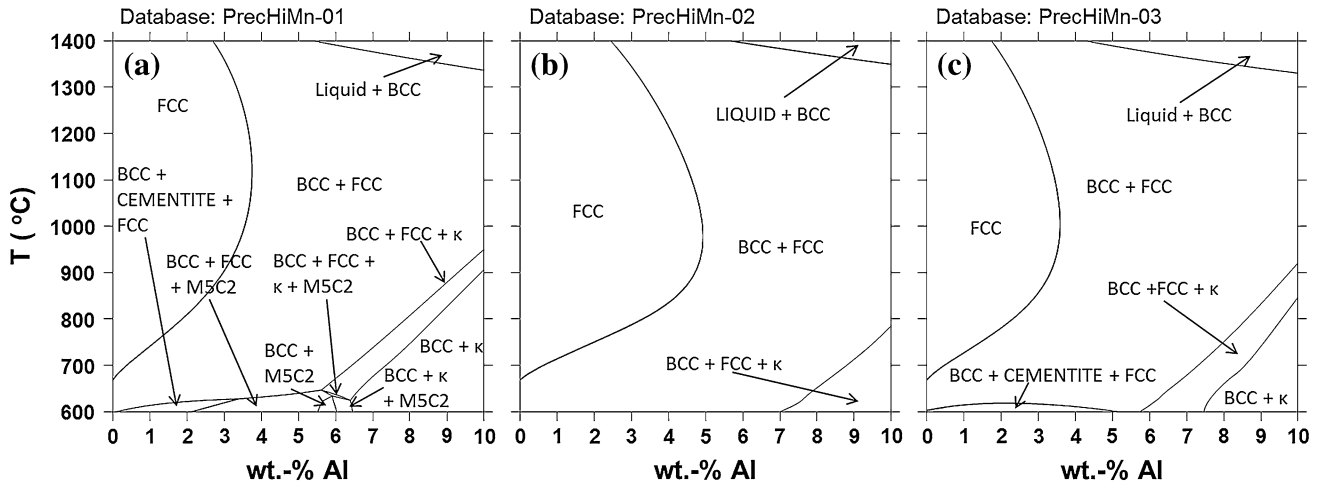


Fig. 2. Isoplethal sections of a quaternary 8Mn-0.2C (wt.%) alloy calculated with three different databases: (a) Hallstedt database incorporating Chin et al. assessment, (b) Hallstedt database incorporating Umino et al. assessment, and (c) Hallstedt database incorporating the assessment by Lindahl and Selleby.

Fe-rich corner, is now well reproduced. Figure 2 shows a comparison of isoplethal sections calculated with these three databases.

In the first database, cementite is replaced at relatively low temperatures by M_5C_2, which is not supported by the experiment. There are differences between the databases regarding the limits of composition of the different phase fields. For instance there is no agreement on the three-phase domain fcc + bcc + κ, which is of primary importance during continuous annealing. Nevertheless, it is believed that the latest assessment should provide the best results. It should be pointed out that the Al-Mn-C system has not yet been evaluated and that the model for the κ-phase needs to be improved. The information provided by these calculations is very important for the design of the composition and process. Figure 3 shows a comparison of austenite phase fractions using again this three databases. Similar results are observed between the first and the third version but are shifted in temperature. Experimental results are now needed in other phase fields than those used for the optimization to validate these new descriptions.

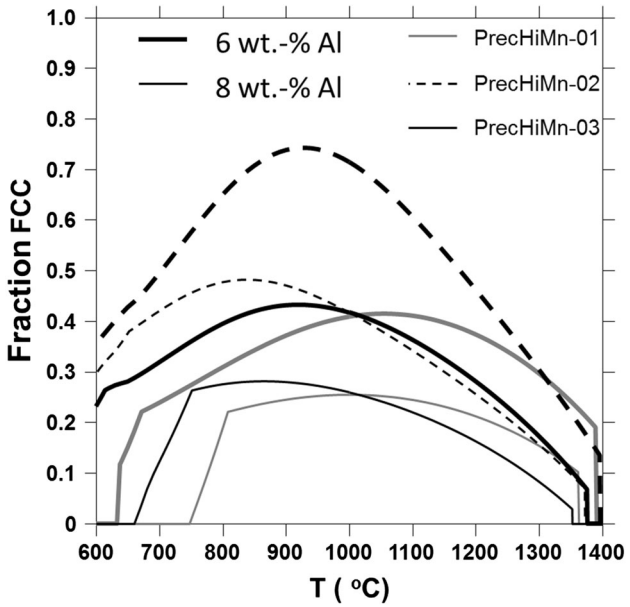


Fig. 3. Austenite fraction as a function of temperature of a 8Mn-0.2C (wt.%) alloy with 6 wt.% and 8 wt.% Al calculated with the three different databases.

MICROSTRUCTURAL ASPECTS OF HIGH-Al STEELS

The isothermal section of a FeMnAlC alloy with 8 wt.% Al at 850°C in Fig. 4 shows the different phase fields occurring by varying C and Mn contents. The bcc phase dissolves a small but not negligible amount of C at lower Mn contents. This phase is still present at higher Mn contents as long as the C content is kept low. Increasing the C content promotes κ-phase formation. In the cases where this phase is not desired, for instance during continuous annealing, a higher Mn content is required to avoid it. To reach a region where fcc is stable at this temperature, high C and Mn contents are required.

Therefore, high-Al steels for automotive applications can be classified in three families: the ferritic alloys that are mainly Fe-Al alloys (Fig. 5a), the two-phase austenite-ferrite alloys in which C and Mn are incorporated in order to stabilize austenite (Fig. 5b, austenite in red), and the austenitic alloys with relatively high levels of C and Mn (Fig. 5c). This last one has been called TRIPLEX because of the additional presence of small amounts of ferrite (Fig. 5c—ferrite in blue) and the intragranular nanometric precipitation of the κ-phase.

Because of the different microstructures and large different in C and Mn, these families present

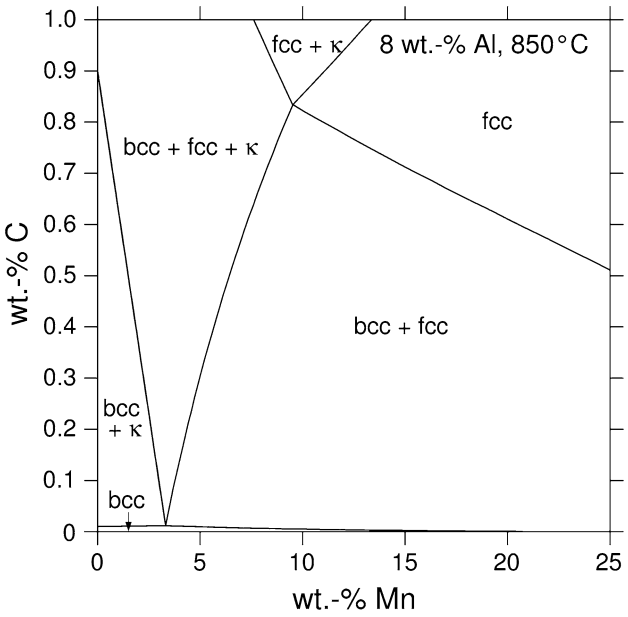


Fig. 4. Iron-rich corner in an isothermal section at 850°C of an 8-wt.% Al showing the phase fields of the three families of lightweight steels described here (bcc, bcc+fcc, and fcc).

different and complex transformation behaviors. Here, some aspects of those behaviors are presented.

Recrystallization of Ferritic Low-Density Steels

Ferritic Fe-Al alloys often exhibit surface defects, referred to as roping, after the deep drawing process. The formation of these defects, which consists of surface corrugations parallel to the rolling direction of the steel sheet, is mainly attributed to the inhomogeneous orientation distribution of individual grains. It is thus important to understand the recrystallization mechanisms to control the microstructure.

Castan et al.[34] studied the effects of strain, strain rate, and temperature on a ferritic Fe-8 wt.% Al alloy using hot torsion. The tests performed in the temperature range from 900°C to 1100°C with a strain rate range of 0.1–10 s^{-1} showed that two types of dynamic recrystallization may operate during hot deformation: At a high temperature and high strain rate, this alloy undergoes discontinuous dynamic recrystallization (DDRX), whereas at a lower temperature and low strain rate, continuous dynamic recrystallization (CDRX) occurs (Fig. 6).

Depending on the deformation conditions, there is a competition between these two mechanisms. To limit the phenomenon of roping, which is mainly attributed to the inhomogeneous distribution of grain orientations, it seems advantageous to promote recrystallized structure according to a mechanism of discontinuous dynamic recrystallization.

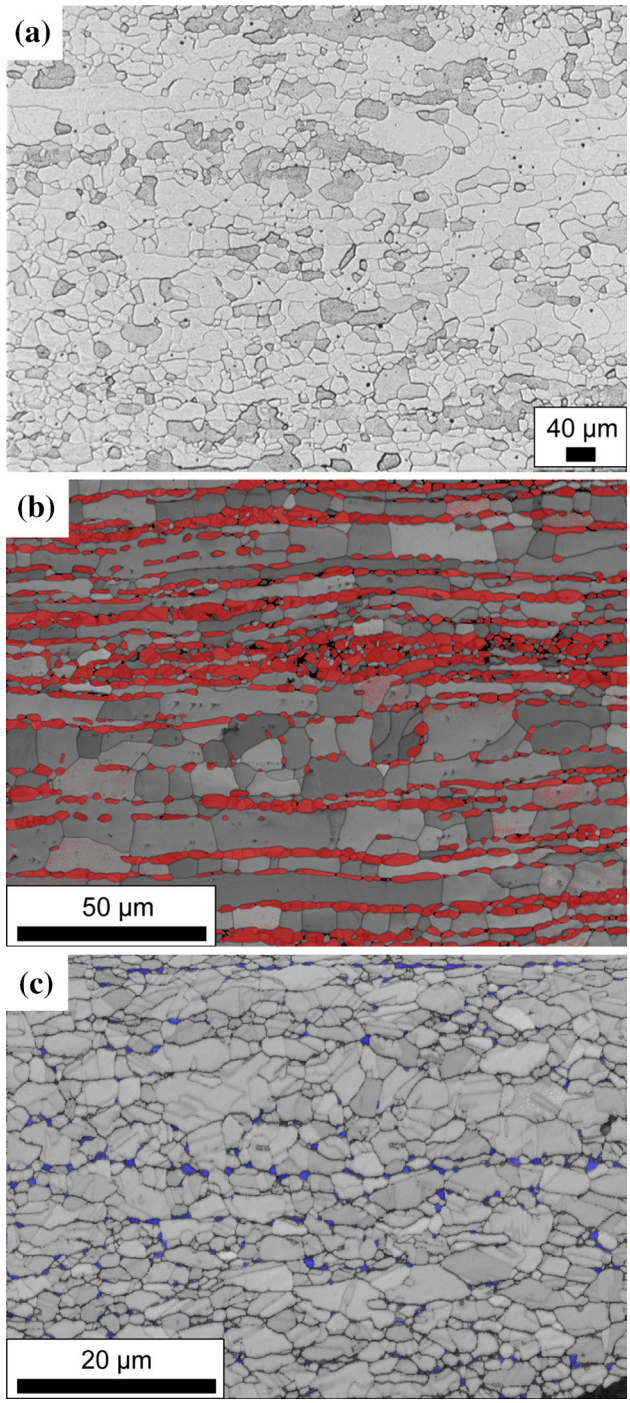


Fig. 5. Typical microstructures of three families of high-Al steels: (a) ferritic, (b) two-phase austenite (red)-ferrite, and (c) three-phase austenite-ferrite (blue). The κ-phase is not observed here due to its size. The last two images are electron backscattered diffraction (EBSD) images.

Complex Microstructures During Cooling of Two-Phase Alloys

At low temperatures, thermodynamics predicts that the austenite in a two-phase structure should decompose to a certain degree into ferrite and the κ-phase (Fig. 2c). A good control of parameters and

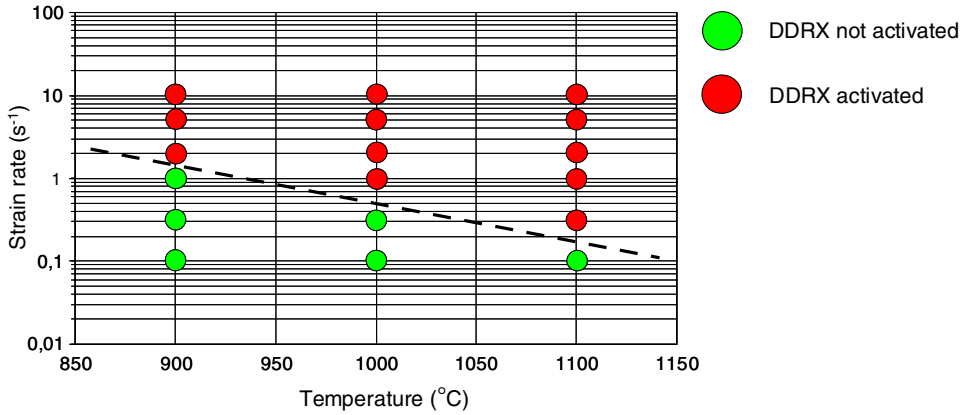


Fig. 6. Strain rate–temperature map displaying the activation domain of the DDRX mechanism in the Fe-8% Al ferritic steel.[34]

choice of chemistry could avoid this keeping a two-phase austenite-ferrite structure at room temperature. To master the final microstructures, it is therefore required to understand austenite decomposition. Zuazo[35] has observed that the precipitation of this phase gives rise to particular microstructures at low Mn contents in which the similarity of crystal structures between austenite and the κ-phase plays a significant role.

For instance, when cooling at low rates from the two-phase domain, the κ-phase precipitates at the interface between austenite and ferrite. Figure 7a shows an example of the resulting microstructure with the κ-phase and ferrite precipitating in a sort of nonlamellar eutectoid. The complexity of the morphology is better observed in three dimensions (Fig. 7b) by using a deep etching procedure. In this case, the κ-phase growths along the long axis of austenite. Two other 3D morphologies of κ-phase precipitation have been observed in this kind of steels using the same technique.[35] At these low Mn contents, isothermal treatments in the two-phase ferrite κ-phase domain promote a complete decomposition of austenite leading to a structure where a sort of cocoon of the κ-phase is left (Fig. 7c).

Evolution of Microstructure in Austenitic Grades

The third low-density steel family, TRIPLEX, has an austenitic matrix due to massive addition of C (up to 1.3%) and Mn (up to 30%Mn). Mn added to stabilize austenite should be less than 35% to avoid the formation of brittle β-Mn.[36] C that contributes also to austenite stabilization is proven to be a strong strengthener for this alloy system.[37–39] As mentioned in the section, "Aluminum: An 'Old' Alloying Element," this grade was initially developed as a possible substitute for Ni-Cr stainless steels.[40] In such applications, aluminum was introduced to increase corrosion and oxidation resistance. More recently, the interest in this family for automotive applications was raised as they exhibit a very interesting strength-elongation compromise evidently combined with a low density. From the three phases that could be present in its microstructure—austenitic matrix, ferrite, and intragranular κ-phases—the precipitation of the nanosized κ-phase inside austenite could lead to important precipitation hardening.

This precipitation has been studied in a 28.8Mn-9.4Al-0.92C (wt.%) alloy. Figure 8 shows the evolution of the microstructure observed by transmission electron microscopy (TEM) during aging at 500°C. The steel was initially solution treated at 1000°C for 10 min and water quenched. Even at this initial state, diffuse superlattice reflections corresponding to the formation of intragranular κ-phases were distinguished on the TEM diffraction pattern of austenite (Fig. 8a). After 2 min of aging, superlattice reflections caused by κ-phases are still weak and diffuse. At this stage of aging, it is difficult to image κ-phases by dark-field imaging. After 16 min of aging, κ-phases are less diffuse and more intense, meaning that their volume fraction has increased (Fig. 8b). After 4 h of aging, they are clearly visible in dark-field imaging (Fig. 8c). They are round in shape and their size is smaller than 5 nm. At the longest time of aging (64 h) used in this study, κ-phases have a cuboidal shape with a size between 5 nm and 10 nm (Fig. 8d). They are arranged along the $\langle 100 \rangle$ directions of austenite.

During such aging, not only austenite can evolve. Indeed, ferrite shows some structure modifications. Ferrite evolution was observed in a 29Mn-9.6Al-0.56C (wt.%) alloy first treated at 850°C for 10 min, water quenched, and then aged at 550°C for 1 h. As can be seen on the TEM diffraction pattern of Fig. 9a, when ferrite is oriented in the [110] zone axis, there are extra spots corresponding to ordered structures. The {111} reflections indicate the presence of the DO_3 structure, whereas the {200} reflections could correspond either to DO_3 or B2 structures. The difference of intensity between {111} reflections and {200} reflections indicates that the initial ferrite structure is a mix of B2- and

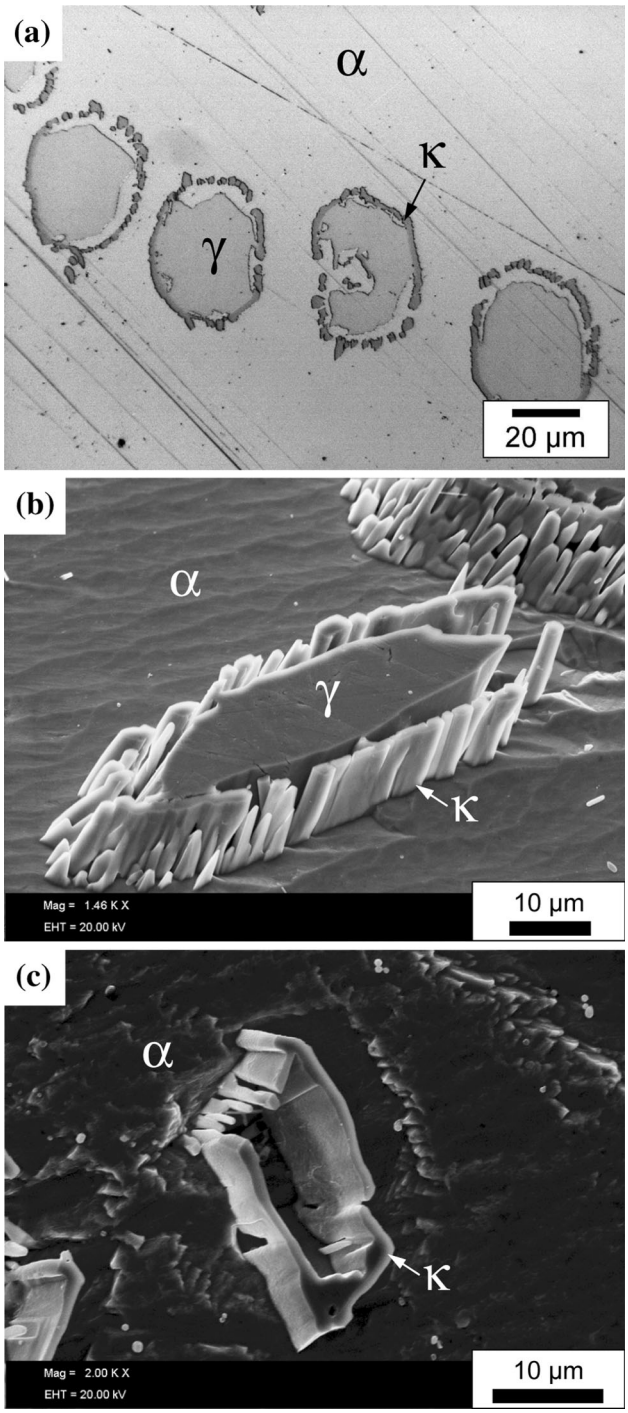


Fig. 7. κ-phase microstructures in a 2Mn-8Al-0.2C (wt.%) alloy: (a) optical micrograph after cooling down from 1400°C at 10°C/s and quenching at 870°C, (b) scanning electron microscopy (SEM) micrograph of (a) but after deep etching, and (c) SEM micrograph after 15 min at 870°C—deep etching.

DO_3-ordered structures. Such ordered structures are not only due to the high level of aluminum of the steel but also due to partitioning between austenite and ferrite that leads to a level of Al in ferrite higher than the nominal composition. After aging for 1 h at 550°C (Fig. 9b), there is no more intensity difference between {111} reflections and {200} reflections on the diffraction pattern; ferrite becomes a fully ordered DO_3 structure.

This section showed some examples of the richness in terms of microstructure and transformation phenomena of low-density steels under the influence of different processing parameters. Finding the right ones is important to obtain the best out of these microstructures in terms of mechanical properties, and the following section intends to improve the understanding of the influence of some of these parameters.

RELATION BETWEEN MICROSTRUCTURE AND PROPERTIES OF LOW-DENSITY STEELS

It has been shown that the additions of Mn and C to a binary FeAl lead to complex microstructures which also present better properties. This is related to the presence of austenite and its decomposition products, as well as to the solid solution effects of these additions. Additional effects come from changes in process parameters that have an impact on phase equilibria and phase morphologies. Here, a couple of examples of the relation between microstructure and properties are shown.

Influence of Annealing Parameters on Two-Phase Microstructures

Figure 10 shows the influence of chemistry and annealing parameters on the tensile strength of austenite plus ferrite microstructures. Increasing the annealing temperature and/or time leads to a decrease in tensile strength with the yield strength behaving in a similar way. One strength of this concept is the observed relative small change in MPa/°C. The influence of Al and Si, both important solid-solution strengtheners, is also observed. Recently, the influence of additions of Al on strain hardening has been studied, showing that Al greatly increases kinematic hardening.[41]

The change in properties seems to be continuous except for the samples treated at 800°C. The observation of the microstructures shows that recrystallization of ferrite is not complete at 800°C in comparison with that at 1000°C (Fig. 11a and b). Ferrite grains in the shape of pancakes are mainly observed at the lower annealing temperatures. Nevertheless, it seems that local misorientations (not shown here) are similar, with inverse pole figure (IPF) maps showing a higher degree of variation in orientations at the lower temperatures. Recovery has probably already taken place to a certain degree.

In the sample annealed at 800°C, austenite is not easily indexed and this could be due either to a transformation of austenite into martensite or to a heavily dislocated austenite. The latter was confirmed by transmission electron microscopy studies.

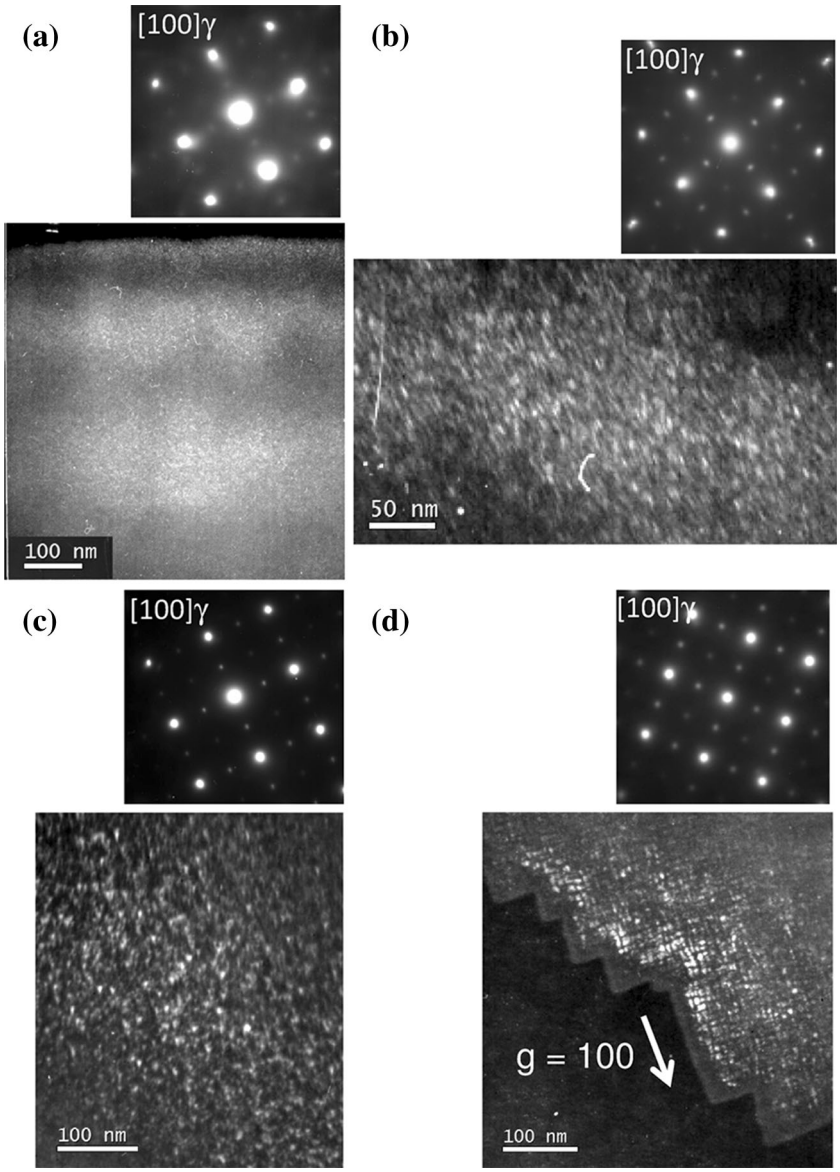


Fig. 8. TEM imaging of microstructure evolution of a 28.8Mn-9.4Al-0.92C (wt.%) during aging at 500°C. For each aging time, the diffraction pattern of austenite in the [001] zone axis and the dark-field image using {100} superlattice reflections are presented: (a) initial stage and 2 min aging, (b) 16 min aging, (c) 4 h aging, and (d) 64 h aging.

The cause of the evolution of properties with temperature and time is certainly due to an influence of recovery and recrystallization of both phases mainly at the lower annealing temperatures and is most likely caused by grain growth at the highest temperatures. Another parameter to take into account is the predicted change in volume fraction of austenite with temperature.

Effect of Microstructure on Properties of Austenitic Grades

A study of the microstructure effects on the mechanical properties of this grade has been recently published by the authors.[38] In that work, thermal treatments have been optimized on four different chemical compositions in order to be able to isolate the effect of different microstructural parameters like grain size, carbon content, and ferrite fraction.

The effect of austenite grain size has been investigated for grain sizes ranging from 1.65 μm to 40 μm in a 28.8Mn-9.4Al-0.92C (wt.%) alloy. A classic Hall–Petch relation was found between (I) grain size and (II) yield strength and tensile strength with a coefficient of 461 MPa $\mu m^{1/2}$ and 351 MPa $\mu m^{1/2}$, respectively. In parallel, uniform elongation and total elongation increase when the grain size increases. However, a saturation effect is observed as soon as the grain size is higher than 10 μm. Figure 12 shows the evolution of the engineering stress–strain curves as a function of austenitic grain size.

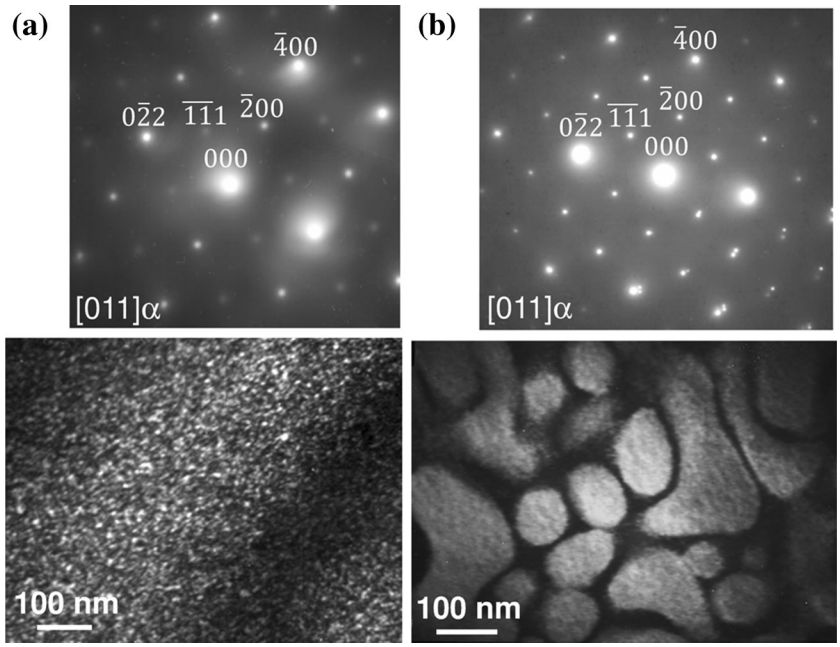


Fig. 9. Ferrite microstructure evolution of a 29Mn-9.6Al-0.56C (wt.%) during aging at 550°C observed by TEM: Diffraction pattern of ferrite in the [011] zone axis and dark-field image using the {200} reflection are presented for (a) the as-quenched sample and (b) after 1 h of aging at 550°C.

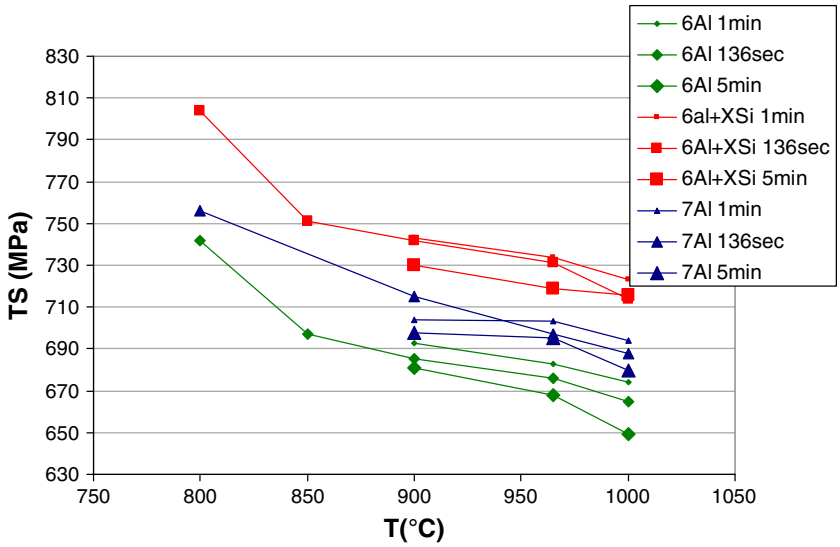


Fig. 10. Influence of annealing temperature, time, and composition content on the tensile strength of a series of 8.5Mn-0.2C (wt.%) austenite-ferrite alloys.

The effect of carbon on yield strength was more difficult to assess as only two modalities were studied, inducing as a consequence, a large uncertainty in the evaluations. Moreover, some κ-phases already present in austenite grains (see the section "Evolution of Microstructure in Austenitic Grades") could have also disturbed the results. Qualitatively, it appears that decreasing the C content leads to a significant decrease in strength. Even if it has been difficult to give a precise quantitative effect of C, the results obtained allowed to considered it lower than 300 MPa/C-wt.% but certainly higher than 187 MPa/C-wt.%.

Finally, the ferrite effect was investigated. A small ferrite fraction (2–5%) increases notably the yield strength by 30–50 MPa. At higher fractions of ferrite, up to 30%, there is no effect on yield strength, but above 30%, increasing the ferrite fraction has a negative effect on yield strength. Increasing the ferrite fraction has a similar effect on

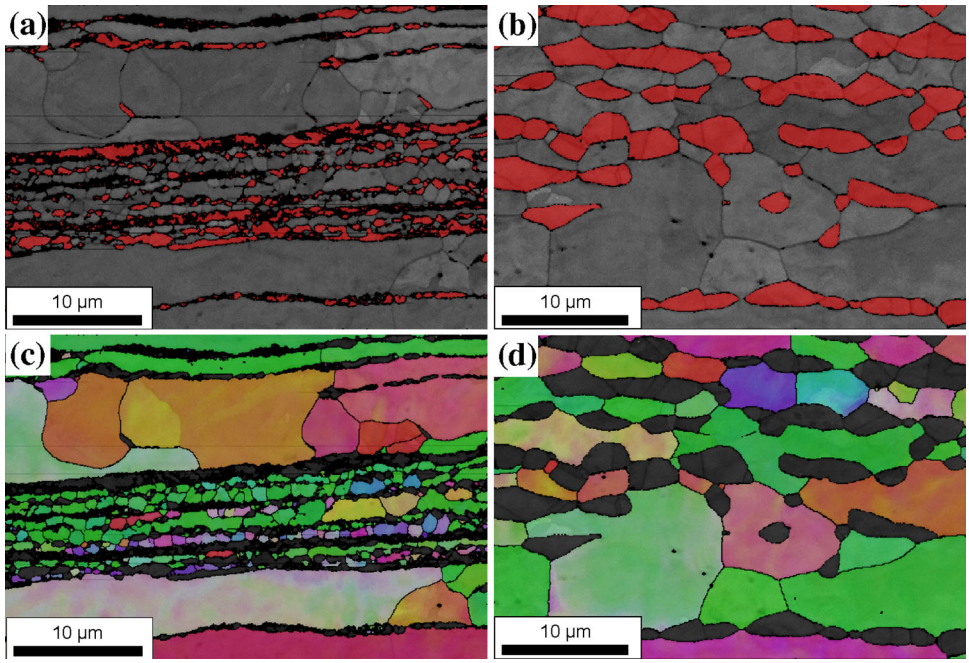


Fig. 11. EBSD phase maps of a 8.5Mn-6Al-XSi-0.2C (wt.%) alloy annealed for 136 s at 800°C: (a) BC + austenite in red and (c) BC + IPF of ferrite and for 136 s at 1000 ºC. At 1000°C: (b) BC + austenite in red and (d) BC + IPF of ferrite. (BC band contrast, IPF inverse pole figure).

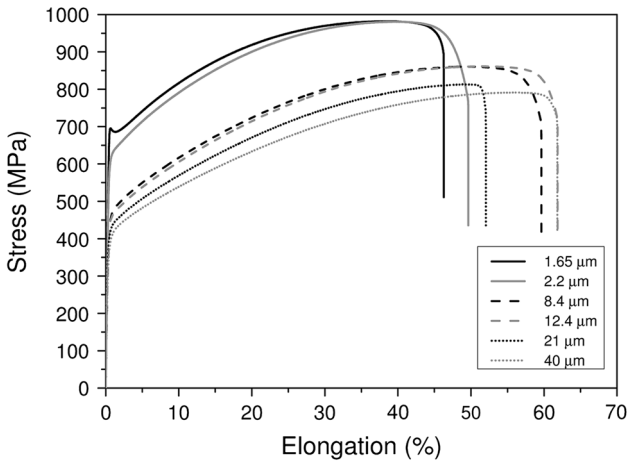


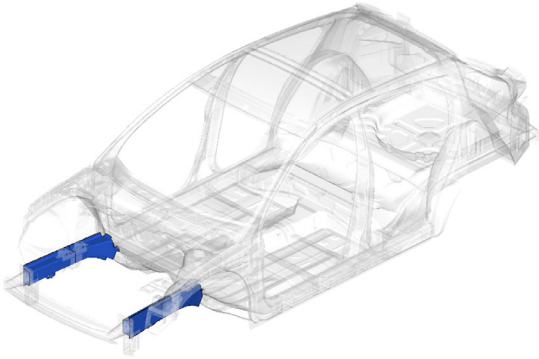
Fig. 12. Engineering stress–strain curves of a fully austenitic 0.92C-28.8Mn-9.4Al (wt.%) alloy for different grain sizes.[38]

ultimate tensile strength and a decrease of around 80 MPa is found when adding up to 39% of ferrite. Regarding elongation, ferrite additions lead to a decrease of ductility. This effect is already important for low ferrite fractions (total elongation decreases 6% with only 3% of ferrite) and is as high as 16% with 39% of ferrite. One must note that the evolutions observed are not only caused by the ferrite fraction itself but most probably represent the global effect of ferrite including the fraction, the morphology (as in the experiments performed ferrite forms in bands), the ferrite grain size, and possibly ordering.

THE POTENTIAL OF LOW-DENSITY STEELS FOR AUTOMOTIVE APPLICATIONS

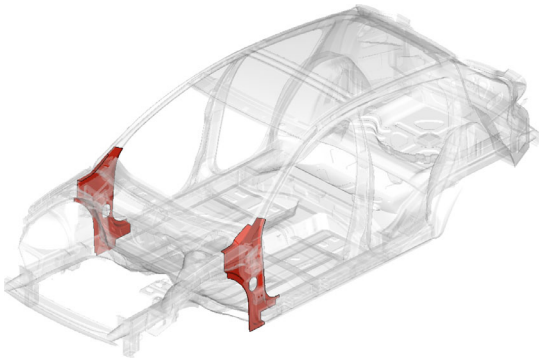
The understanding of microstructural evolution shown in previous sections permits tailoring of in-use properties. These properties, together with the reduction in density, can be used to calculate the weight savings potential provided by these alloys on automotive parts. For this purpose, finite-element front crash simulations of a full vehicle were considered (EuroNCAP 40% overlap front crash). The simulations were done using LS-DYNA finite-element code (Livermore Software Technology Corporation, Livermore, CA, USA), and the materials were modeled by piecewise linear plasticity formulations (MAT_024[42]) based on experimental tensile data. Two automotive parts made of DP600 were used as references and compared with the same parts but made of austenite-ferrite (7.4Al-8.2Mn-0.2C wt.%) (YS: 646 MPa, UTS: 800 MPa, TEL: 26%) and fully austenitic (9.3Al-25.8Mn-0.96C wt.%) (YS: 812 MPa, UTS: 1027 MPa, TEL: 32%) low-density alloys. A front-side member (energy absorption) and A-Pillar (anti-intrusion) parts were chosen as reference components because of their different functions in the structure (Fig. 13). For each part, the thicknesses of the low density alloys were adapted to match DP600 reference performances in terms of energy absorption and section forces. The results show the important lightening potential of the lightweight alloys, around 20% for austenite-ferrite and 30% for austenitic alloys, in comparison with DP600. The potential of austenitic alloys is larger because of its better combination of

(a)



		DP600 Ref.	FeAl $\alpha + \gamma$	FeAl γ
Thickness Fr / Rr (mm)		1.7 / 2.1	1.5 / 1.8	1.3 / 1.7
Total Mass / vehicle (kg)		5.88	4.63	3.99
Weight saving / vehicle	kg	-	1.25	1.89
	%	-	21 %	32 %

(b)



		DP600 Ref.	FeAl $\alpha + \gamma$	FeAl γ
Thickness (mm)		1.6	1.4	1.25
Total Mass / vehicle (kg)		6.11	4.84	4.12
Weight saving / vehicle	kg	-	1.27	1.99
	%	-	21 %	33 %

Fig. 13. Calculated weight-saving potential of austenite-ferrite and austenitic lightweight steels versus DP600 for (a) front side member and (b) A-Pillar.

resistance and elongation plus a larger density reduction. In addition, cost estimations (€/kg saved) of austenite-ferrite, not shown here, showed that in spite of the alloying additions, the €/kg saved is similar to other less alloyed concepts.

SUMMARY

In this overview, the complexity of introducing high-aluminum additions has been highlighted from the point of view of thermodynamics, microstructure development, and properties. An important step forward has been taken with the introduction of high aluminum contents in the frame of CALPHAD-type modeling leading to new thermodynamic databases. The evolution of the microstructure is complex, but a good understanding has been reached. It shows that there is still room for microstructure improvement and better mechanical properties. In their actual state, the combination of properties and density reductions in lightweight alloys shows an important potential for automotive applications. All this allows proceeding with a larger risk assessment concerning upstream and downstream processing for industrial applications.

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